



DELHI PUBLIC SCHOOL NEWTOWN

SESSION 2020-21

MONDAY TEST

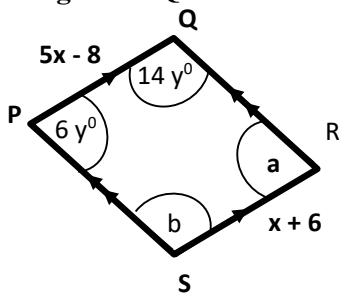
CLASS: IX
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 07. 12. 2020

GENERAL INSTRUCTIONS:

- The paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

1. Two poles of heights 6m and 11m stand on a plane ground. If the distance between their feet is 12m, find the distance between their tops. [4]
2. In a quadrilateral ABCD, angle B = 90° = angle D. Prove that:
 $2AC^2 - BC^2 = AB^2 + AD^2 + DC^2$. [4]
3. ABC is an equilateral triangle of side $2x$. Prove that its altitude is $\sqrt{3}x$. Also find its area. [4]
4. A bicycle wheel of radius 14cm is making 25 revolutions in 12secs. Find the speed of the bicycle in km/h. [4]
5. Find the values of 'x', 'y', 'a' and 'b', with reasons, in the following figure where PQRS is a parallelogram. $PQ = 5x - 8$ and $RS = x + 6$. [4]



6. ABCD is a parallelogram in which $AB = 2AD$ and P is the midpoint of AB. Prove that angle DPC = 90° [5]
7. In a triangle ABC, the internal bisectors of angle B and angle C meet at O. Prove that OA is also internal bisector of angle A. [5]

8. ABC and DBC are two isosceles triangles on the same base BC and vertices A and D are on the same side of BC. If AD is extended to intersect BC at P, show that:

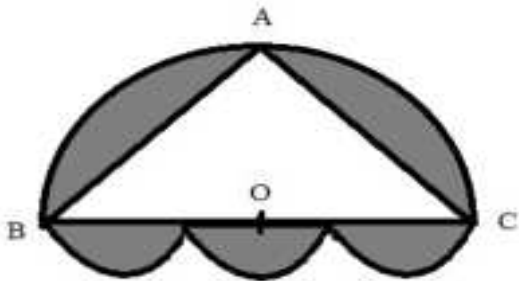
i) $\triangle ABD \cong \triangle ACD$

ii) $\triangle ABP \cong \triangle ACP$



[5]

9. A doorway is decorated as shown in the figure given below. There are 4 semi-circles. BC, the diameter of the larger semicircle is of length 84cm. The centres of the 3 equal semi-circles lie on BC. ABC is an isosceles triangle with $AB = AC$. If $BO = OC$, find the area of the shaded region. [Take $\pi = \frac{22}{7}$]



[5]